SimWorld Research White Paper

A reproducible reduced model and a research program for emergence across physical and social scales

29 October 2026 | Version 1.0 | Implementation baseline physics_build 0.1

Abstract

SimWorld proposes a continuous simulated history in which higher-level structures arise from local interactions, extending from a hot physical seed through chemistry, planetary environments, life, technology and institutions. This paper formalizes that ambition as a multiscale research program and assesses the executable baseline against its design documents. The present implementation is a small stochastic model of thermal populations, effective nucleon formation, light-nucleus reactions and electron capture. It includes event logging, deterministic re-execution, canonical state hashes and restricted skipping of quiet cells. These are concrete foundations for reproducible simulation experiments.

An audit of the 29 October 2026 source snapshot found that all 65 existing tests passed. A fresh eight-cell, 3,400-tick realization at seed 42 reproduced the archived summary and decompressed event log, and replay matched all 34 recorded hashes. Its helium-4 mass fraction was 0.2276 and its reported ionized fraction was 0.003457. These results establish repeatability within the tested environment and satisfaction of project acceptance bands; they do not establish quantitative agreement with cosmology or a derivation of matter from fundamental quantum dynamics.

Source inspection and targeted diagnostics identify unresolved reverse-reaction behavior, an absorbing fully neutral atomic state, repeated random-stream prefixes, and a mismatch between prototype performance measurements and production storage. The roadmap also requires several unimplemented interfaces and closure models. We therefore distinguish implemented behavior, archived prototype measurements, proposed mechanisms and new validation recommendations. The resulting program prioritizes reversible kinetics, explicit uncertainty, independent benchmarks and tested transfer of information between scales. Its central research question is whether a hierarchy of transparent effective models can preserve useful causal structure while supporting increasingly complex, unscripted behavior at affordable computational cost.

Keywords: multiscale simulation; emergence; reproducibility; stochastic kinetics; coarse-graining; artificial life; agent-based modeling.

Evidence date: 29 October 2026. The accompanying research materials contain the editable manuscript, source snapshot, SHA-256 manifest, fresh run, test report and audit diagnostics.

1 Research problem and scientific position

The project asks whether one causal history can connect microscopic interactions to macroscopic organization without scripting the order of discoveries or prescribing a final society. The roadmap emphasizes computed properties, regenerable caches, persistent history and observation-dependent allocation of detail. Its 17 phases supply a wide research scope, while BUILD0 specifies a deliberately narrow first executable slice. A scientifically useful account must preserve both the ambition and the limited domain of the evidence. [S1, S2]

The defensible organizing hypothesis is that higher-level models can retain selected consequences of lower-level dynamics through explicit interfaces. Such models necessarily choose state variables, effective rates, constitutive relations and learning rules. A property computed from a fitted formula is derived within that model, but it is not an independent first-principles prediction. Avoiding element names or runtime lookup tables does not remove calibration, structural priors or excluded mechanisms. SimWorld should disclose these choices as part of the model rather than treat them as violations of its purpose.

Relationship to prior research

Precision big-bang nucleosynthesis provides a benchmark for coupled weak and nuclear reaction calculations, including uncertainty treatment. The observational deuterium abundance provides an independent target with a scale far below the granularity of this baseline. Recombination research also demonstrates that expensive microscopic calculations can be compressed into effective rates while retaining a carefully defined physical domain. These precedents support derived caches and model reduction, provided their accuracy is measured. [1-3]

Heterogeneous multiscale methods couple coarse evolution to local microscopic calculations. They motivate the proposed hierarchy, but do not prove scale separation for the entire chain from particles to institutions. For stochastic reactions, Gillespie's algorithm supplies a reference sampler for a specified well-mixed kinetic model. For agents, the ODD protocol supplies a disciplined description of purpose, state, scheduling and submodels. Each addresses a distinct part of scientific reproducibility. [6-8]

Claims that the present evidence can support

The present contribution is an auditable reduced simulation and a structured research agenda. This paper makes no claim that SimWorld solves quantum field theory, predicts primordial abundances precisely, produces general chemistry, generates life, or explains human history. Those questions require additional models and experiments. The roadmap's unbuilt phases are evaluated as hypotheses about mechanisms and interfaces, not as achieved capabilities.

A meaningful emergence claim names the local rules, the measured higher-level pattern, the initialization and the interventions that distinguish competing explanations. Successful replay shows that an experiment can be repeated. Independent validation asks whether the experiment represents its intended physical or social system. A useful model can satisfy one criterion before it satisfies the other.

2 Evidence and audit methods

The evidence base consists of the roadmap, all detailed phase documents available for phases 0 through 12, the BUILD0 contract, the frozen specification records, executable Python modules, tests, two prototype studies and archived runs. Phases 13 through 16 have roadmap entries without corresponding detailed phase files. Source-level assertions were checked against executable paths rather than accepted from comments alone. Literature selection was focused on the methodological issues raised by the project; it was not a systematic review. [S1-S7]

Evidence class	Material examined	Permitted interpretation
Design intent	ROADMAP and PHASE0 to PHASE12; BUILD0	Proposed scope, assumptions and acceptance criteria
Implemented mechanism	sim package and tests	Behavior supported by inspected code and executed checks
Archived measurement	M3 and M4 run records; spike reports	Historical observation with its own workload and environment
Fresh verification	65-test suite; seed-42 run and replay; targeted diagnostics	Reproduction and counterexamples in one tested environment
External context	Primary research and methodological papers [1-10]	Benchmarks and methods; no endorsement of SimWorld

Table 1. Evidence categories prevent proposed mechanisms and prototype measurements from being reported as capabilities of the main engine.

The full suite was executed with Python 3.11.9, NumPy 2.4.6 and pytest 9.1.1 on the available Windows host. All 65 tests passed in 123.64 seconds. The recorded nightly XML in the original folder covered a 13-test subset; it was not used as a substitute for this fresh full-suite run. The fresh demonstration used seed 42, eight cells, 3,400 ticks, probe cells 2 and 3, and default jump behavior. [S5, S7]

Verification compared complete summary objects, saved hash sequences and SHA-256 digests of decompressed event logs. Comparing decompressed logs avoids treating gzip container metadata as simulated history. The fresh run and replay took approximately 5.55 seconds each while the test suite was also active. These are single observations under concurrent load, not controlled performance benchmarks or estimates of future scalability.

Three additional diagnostic cases tested mechanisms outside the existing acceptance suite: dissociation from a product-only nuclear population, evolution from a fully neutral atomic population, and repeated calls to the same random-stream coordinates with different payloads. The exact diagnostic program and output accompany the paper. These checks expose limitations even though the current tests pass. They do not replace a comprehensive statistical or physical validation campaign. [S7]

No simulation source was modified for this paper. A file-level SHA-256 manifest records the inspected implementation and design snapshot. The local build string is retained for compatibility, but content hashes are necessary because a version label alone does not identify every source change.

3 The implemented reduced physical model

State and expansion

The executable world contains cell temperatures, quant records, integer nuclide populations, captured-electron counts and summary statistics. QuantStore uses Python dictionaries of per-quant lists and per-cell species pools, with NumPy count arrays. It does not implement the fully array-based structure-of-arrays layout described in the frozen D1 decision. Nuclide and atomic states are aggregate counts. The current cell system has a shared analytic temperature and does not evolve spatial fluid transport, gravity or a three-dimensional universe. [S3, S4]

$$t_n = 10^{-5}\,(1.01)^n \text{ s}, \quad a(t) = (t / t_0)^{1/2}, \quad T(t) = 100 / a(t) \text{ MeV}. \quad (1)$$

Equation 1 prescribes radiation-like cooling throughout the run. It is a background schedule rather than a Friedmann evolution solved from simulated energy density. The five effective field entries carry specified masses, charges, spins and coupling labels. Thermal target counts are rounded functions proportional to T cubed with a piecewise massive-particle suppression factor, scaled by the calibration volume. Thus the existence and properties of these field entries are inputs. [S4: constants, fields, lattice and condense]

Effective nuclei and reaction discovery

Weak flavor conversion supplies a frozen neutron probability. Below the confinement threshold, available quanta are consumed in triples, and neutron/proton labels are sampled from that probability. The consumed triple need not carry the sampled label's exact flavor content; the project records this as sea completion. Nuclear binding then uses a fitted liquid-drop expression, with nucleons assigned zero binding:

$$B(Z,A) = a_v A - a_s A^{2/3} - a_c Z(Z-1)/A^{1/3} - a_a(A-2Z)^2/A + a_p \delta(Z,A) A^{-3/4}. \quad (2)$$

The pairing factor is zero for odd A and has opposite signs for even-A systems with even or odd Z. Five fitted coefficients are published in constants.py. The network considers a bounded set with Z at most 4 and A at most 12. A candidate sum must have positive computed reaction energy, but additional explicit rules reject mass-5 nuclei and products with neutron-proton imbalance greater than one. These structural restrictions are model assumptions; the resulting stability pattern is not wholly emergent from the binding formula. [S4: nucbind, bind and network]

Forward reactions use population products, a fitted rate scale and charged-particle suppression. Rates are converted to capped per-tick probabilities, and the fixed update order limits which events can occur in each tick. This is not Gillespie sampling, and its rates cannot be interpreted as continuous-time physical rates without a consistent time mapping and convergence study. The reverse-channel implementation has additional limitations discussed in Section 6. [8, S4]

4 Atoms observation and reproducible execution

Electron capture

The atomic module computes the hydrogenic scale $I = \alpha_{em}^2 m_e/2$, approximately 13.61 eV for the configured constants. It stores one captured-electron count per cell and nuclide, using a spread-filling rule to reconstruct capture levels. Free electrons are the difference between total nuclear charge and captured electrons. This enforces a charge-accounting identity; it does not simulate electrons as orbital wavefunctions or a dynamical photon bath. Capture is disabled above $T_{gate} = 3I$. [S4: atoms]

The comparison function is a model-specific Saha-like solution. Writing $L = \exp(I/T)/\lambda^3$ and $K = NL$, it returns the positive solution to $Kx^2 + x - 1 = 0$:

$$x_S = [\mathrm{sqrt}(1 + 4K) - 1] / (2K). \quad (3)$$

Here N is the hydrogen count per cell, $\lambda^3 = (\mu T\ \mathrm{SAHA_PHI_E})^{3/2}$, and $\mu = m_e A/(A+1)$, with A = 1 for hydrogen. The phase-space factor is calibrated in engine units. Equation 3 is an internal reference, not independent cosmological data. Fully captured species are excluded from updates, so the kernel need not relax to this equilibrium from every initial condition. Cosmological recombination requires richer atomic and radiative physics. [3, S3, S4]

Observation and quantum terminology

Quanta move between records named superposed and collapsed when probe coverage or a threshold interaction rule triggers materialization. The current representation does not evolve a general many-body wavefunction or density matrix. Environmental decoherence concerns suppression of interference under interactions; objective-collapse theories introduce specific testable changes to dynamics. A software threshold is not a derivation of either process. In this baseline, collapse is best treated as an effective state transition. [4, 5]

The roadmap's original perception-centric wording is narrowed by BUILD0: perception should govern detail rather than existence. Yet probe-triggered collapse also removes quark rows from weak conversion in covered cells. Existing probe-equivalence checks are limited to selected thermal-population and removal observables. Full observer independence of the baseline is therefore unestablished. Physical measurement interventions, numerical refinement and display visibility need separate contracts. [S2-S5]

Replay and skipping

Random values come from a deterministic counter-based function. EventContext records event keys and payload digests; replay checks event order, periodic state hashes and the final hash. These hashes are not cryptographically linked. The evidence supports same-machine, same-build reproduction. Seeded pseudorandom draws do not establish intrinsic quantum randomness; the implementation does not ingest an external quantum entropy record. Quiet-cell jumps match reference evolution in selected tests. That observed equality does not prove equivalence for arbitrary runs or future coarse-graining. [S4, S5]

5 Observed results and their interpretation

The fresh seed-42 realization reproduced the archived M4 summary exactly. The final state hash and all 34 hashes matched during replay. Table 2 reports model outputs with enough precision for comparison, while distinguishing acceptance criteria from empirical validation. The experiment is one realization, not an ensemble estimate. [S6, S7]

Observable	Fresh result	Interpretation
Initial settings	8 cells; 3,400 ticks; seed 42	Same configuration as archived M4
Neutron/proton ratio at freeze-out	0.129890	Inside internal band 0.10 to 0.20
Tracked freeze-out temperature	1.08087 MeV	Model-dependent diagnostic
Helium-4 mass fraction Yp	0.227618	Inside internal band 0.20 to 0.30
Hydrogen mass fraction XH	0.747344	Inside internal band 0.69 to 0.80
D/H and Li/H	Both zero	D/H admits zero; Li/H is record-only
Reported ionized fraction Xe	0.00345722	Charge-based electron ledger statistic
Reported neutral fraction	0.996543	Bound-electron fraction 1 - Xe; not neutral-atom number fraction
Nucleons formed; nuclear reactions	1,318; 247	Small finite-population experiment
Jump events; skipped cell ticks	6; 6,000	Cell work omitted, not 6,000 global ticks
Final model time; temperature	4.87921×10^{9} s; 4.52715×10^{-6} MeV	Approximately 155 years under Eq. 1

Table 2. Fresh run values from audit_diagnostics.json and fresh_m4/summary.json. Full precision is preserved in the supplementary data.

Pitrou and colleagues reported $Y_p = 0.24709 \pm 0.00017$ for their precision calculation. The baseline's value is about 7.9% lower in relative terms. This comparison supplies scale, not a statistical significance: the models do not share a calibrated cosmological specification or uncertainty budget. SimWorld's broad internal band is a regression criterion. [1]

Cooke and colleagues measured $D/H = (2.527 \pm 0.030) \times 10^{-5}$ in a seven-system analysis. With only roughly 10^3 hydrogen nuclei, the expected deuterium count at that abundance is about 0.03. A zero count is unsurprising and cannot test that abundance. An ideal Poisson count near 100, needed for roughly 10% counting precision alone, would require about four million hydrogen nuclei. This is a sampling-scale estimate, not a proposed universal particle threshold. [2]

6 Validation gaps exposed by the audit

Passing tests establish the properties they assert. The suite includes useful replay, tamper detection, conservation, cache and acceptance checks, but several stronger statements in comments and specifications do not follow from the executed kernels. The following findings limit the scientific interpretation of the outputs. They are documented without modifying the engine. [S4, S5, S7]

Finding	Evidence	Consequence
Reverse nuclear pathways depend on forward populations	network._rates skips a pair when either reactant is absent. A product-only diagnostic omitted the p+n reverse channel.	Products cannot dissociate through that channel until the required reactants already exist.
Reverse rates do not implement the stated channel ratio	The reverse expression uses product binding B rather than reaction Q and lacks the forward Gamow factor.	The documented detailed-balance claim needs a derivation and equilibrium tests.
Fully neutral atomic populations are absorbing	100 neutral H nuclei at Tgate/1.2 produced zero events over 1,000 ticks; the internal Saha function gave Xe = 0.10815.	The kernel does not relax from this state to its own nonzero ionization reference.
Random draws reuse stream prefixes	Calls sharing seed, tick, kind, cell and sub restart at the same counter despite different payloads.	Repeated species calls can be correlated; reproducibility does not imply independence.
Storage differs from the performance prototype	Main quant records are dictionaries and lists; Spike A uses array operations.	The prototype throughput cannot be attributed to the current engine.

Table 3. Source findings and targeted diagnostics. The nuclear test allowed other channels to react; the finding is specifically the missing dissociation pathway, not an absence of all reactions.

For closed systems intended to relax to thermal equilibrium, the kernel must admit the correct equilibrium distribution and the intended reverse pathways. Microscopic detailed balance requires $\pi(x)q(x\to y) = \pi(y)q(y\to x)$ at equilibrium π for each reversible transition pair. Open driven systems need not satisfy global detailed balance. A paired rate formula is insufficient if enumeration removes the reverse move or scheduling changes its effective probability. These issues need resolution before the yield bands support physical claims.

The project's own surprises ledger also records imposed baryon excess, flavor bookkeeping approximations, unresolved deuterium and lithium behavior, and recombination much earlier than the physical epoch. Photon and entropy dynamics remain inadequate for a quantitative recombination claim. ETA and EPS_B are declared constants but do not actively control the run; the quark excess is preserved through removal exemptions. Parameters must be traced to operative code before they enter sensitivity studies. [S3, S4]

7 Prototype results and computational feasibility

Two prototype studies inform the roadmap, but their evidence has a different scope from the integrated run. Spike A measures operations on array storage; Spike B tests a fitted approximation to first ionization energies. Both are useful feasibility probes when their workload and calibration are stated. Neither demonstrates a scalable universe simulator. [S6]

Prototype	Recorded result	Limit of inference
A quant storage and update cost	281.4 million quanta/s for an evolve kernel; 20.4 million interactions/s; 13.899 ms mean full tick	One synthetic workload on the recorded host. Main QuantStore has a different representation.
B shell approximation	41 training examples; fitted screening; 6.9% training RMS relative error	Empirical fit, not a self-consistent-field atomic solver or general chemical model.
B selected probe elements	Reported errors: He -16.9%, Li +4.4%, C -18.2%, Fe -4.1%	Small probe set; H uses an analytic anchor. No broad out-of-distribution or molecular validation.

Table 4. Archived prototype measurements from spikes/a_quant_cost/report.json and spikes/b_shells/report.json. No new prototype timing benchmark was performed.

The current main engine allocates and updates Python objects in paths that the original cost model expected to be array operations. A future storage refactor can be justified by a measured bottleneck and checked by replay or appropriately defined statistical equivalence. It must then be benchmarked as an integrated workload including reaction discovery, event logging, hashing, memory traffic and observation changes.

The proposed scale targets include a four-billion-year planetary run with roughly 10,000 resolved elements, a compressed 500-million-year biological era, and 10,000 compressed social agents with at most 1,000 instantiated. These are unmeasured planning targets; future-phase performance tables contain no measurements. Raw quanta per second cannot price chemical discovery, sparse ecological change, long memory or a sudden request for high detail. Each scale needs a workload model with peak memory, log growth and reconstruction cost. [S1: PHASE7, PHASE8 and PHASE11]

Derived caches are compatible with the research objective when their provenance, domain and error are known. Effective multi-level recombination is an example of reducing recurring computation by separating atomic calculations from cosmological evolution. Its success in a defined problem does not license arbitrary compression of chemical or social histories. [3]

A useful runtime report should show a fidelity-versus-cost curve over multiple problem sizes and seeds, plus the worst observed transition into detailed simulation. Hardware, implementation versions, warm-up, repetitions and variability belong alongside throughput. Until those measurements exist, CPU/GPU partitioning and civilization-scale capacity remain design proposals.

8 Architecture for a multiscale research program

The strongest architectural idea in the roadmap is a persistent causal record shared across levels. Bound objects, materials, organisms and institutions should carry enough provenance to explain their creation and subsequent influence. Each transition must define which information is retained, what is approximated and which future queries remain answerable. Figure 1 separates the present reduced model from proposed layers. [S1-S4]

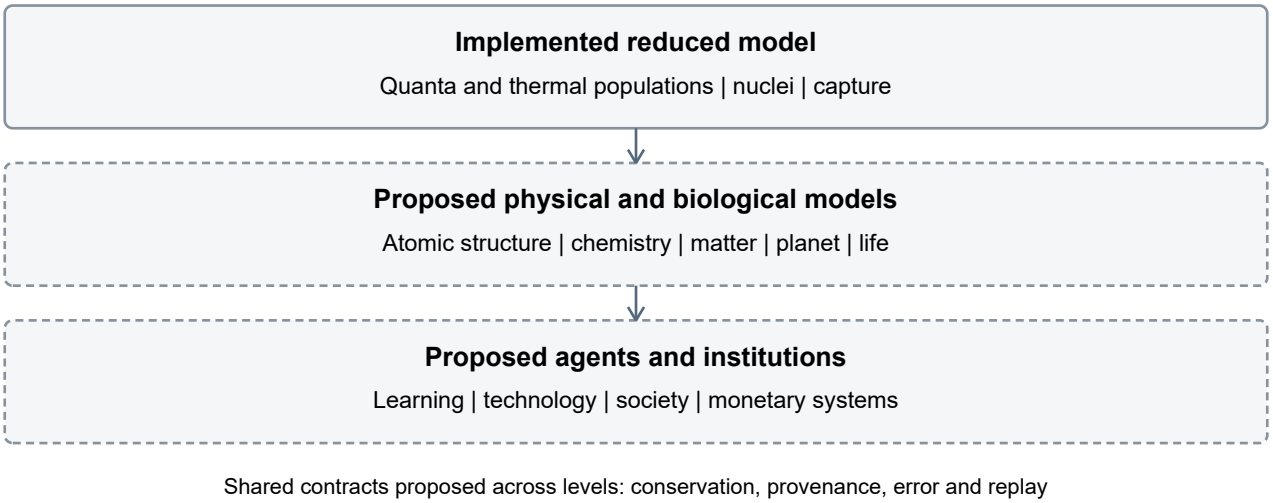


Figure 1. Scope of the current implementation and the proposed hierarchy. Solid outline denotes executable baseline scope; dashed outlines denote future systems. Arrows indicate intended dependencies, not demonstrated end-to-end emergence.

Four contracts are needed at each interface. A **state contract** declares units, conserved quantities, sufficient statistics and boundary inputs. An **event contract** specifies causality, scheduling, random-stream identity and attribution. An **error contract** identifies discarded information and the observable error introduced by compression. A **reconstruction contract** states how detail is regenerated and which histories remain compatible with the summary. These are recommendations that extend the existing interface discipline.

For a fine state x, a restriction R(x) produces a coarse state and a lifting L reconstructs compatible detail. Exact equality L(R(x)) = x is generally impossible after information is discarded. Instead, the model should identify an observable O and assess |E[O(x)] - E[O(L(R(x)))]| against a declared tolerance, while separately testing conservation and distributional differences. Exact replay of the chosen coarse algorithm is a different requirement. [6]

Conservation does not mean that every quantity is closed in every experiment. Expansion, radiation, food, external stress or migration can cross boundaries. The contract must expose those fluxes so a budget closes over state changes plus transfers. Ownership records, knowledge and money add informational claims about physical resources; they must not silently create matter or energy. [S1: PHASE5-PHASE12]

Observation should request refinement only when the retained state can support it. Unseen regions can still affect future events through radiation, transport or agents. Omitting these influences because they are off-screen would change the modeled system. This distinction belongs in the earliest architecture, even though the roadmap assigns perception and runtime their own later phases.

8 Architecture continued

Feedback experiments across levels

The proposed world is more than a sequence of increasingly complicated objects. Later systems must alter the conditions that support them. Three cross-layer experiments illustrate this intended feedback and provide narrower research targets than an entire history. None is implemented in the current micro-seed. [S1: PHASE8-PHASE12]

Ecology and planetary reservoirs. A population consumes substrates, releases products and changes local nutrient and oxygen availability. Transport redistributes those changes, which then alter the population's growth and competitors. Compare a fully coupled system with one in which environmental feedback is held fixed while the same external energy enters both. Measure reservoir closure, population persistence and the response to a resource shock. This tests whether ecological effects are carried through the physical interface rather than assigned as a global vegetation bonus.

Cooking materials and learned behavior. An agent changes a material through heating, and the resulting chemistry changes usable energy or tool performance. Those returns influence future choices, while fuel consumption changes the environment. A controlled experiment can hold the sensory and learning model fixed and vary access to the physical transformation. Observable outcomes include net energy after fuel and processing costs, transfer of a learned procedure to unfamiliar resources, and persistence of that procedure across agents. A successful result would support one coupled mechanism; it would not establish an inevitable route to civilization.

Industry exposure and collective maintenance. Production moves material and releases pollutants; transport determines exposure; agents perceive consequences and decide whether to maintain, regulate or abandon an asset. Compare matched communities under different communication reliability and maintenance costs. Measure exposure, resource use, repair effort and the distribution of outcomes. Ownership and obligation records can attribute choices and costs, while the physical model determines where the pollutants go. Political labels should be assigned only after the behavior is characterized.

What compressed histories must retain

The biological plans retain genotype frequencies and host-parasite correlations. The cognitive and social plans add skill and role distributions, ancestry and correlations between lineages and culture. These joint relationships matter: independently preserving a skill histogram and a resource histogram can lose which agents can use which resources. Stable periods may permit large temporal jumps, while shocks or strong interactions require detailed evolution. [S1: PHASE8-PHASE11]

Tests should reconstruct detailed populations from a summary and compare response distributions to matched shocks. The closure is inadequate if identical coarse records hide histories that predict materially different futures for the chosen observables. Retaining more correlations may improve fidelity at a memory and computational cost. Measuring that tradeoff is a central contribution the project could make, even if the full end-to-end history remains out of reach.

9 Roadmap from world rules to continuum physics

Table 5 translates phases 0 through 6 into research deliverables, combining selected source gates with the validation requirements developed in this paper. They are proposed requirements unless identified as demonstrated earlier. The micro-seed spans simplified nuclear and atomic behavior within the Phase 0 build; this does not mean that the full Phase 2 or Phase 3 programs are complete. [S1, S2]

Phase and intended mechanism	Required evidence	Present status and main risk
0 World rules Quant records, fields, time, observation and replay	Frozen state/event contracts; runnable cooling seed; measured cost and replay	Reduced baseline exists. Physical interpretations exceed what representation tests establish.
1 Quantum layer Array storage, field kernels, interference, entanglement and scheduling	Interference controls; non-signalling statistics; spectra; reproducibility and throughput	Detailed proposal. Correlated draws or threshold collapse alone are insufficient.
2 Nuclear layer Residual forces, binding, reactions, decay and energy deposition	Held-out binding curves; light nuclei; stellar yields; event conservation; decay closure	Detailed proposal beyond toy kernel. Fitted anchors must be separated from predictions.
3 Atomic layer Radial self-consistent fields and correction hierarchy	Convergence; ionization and affinity; spectra; high-Z limits; molecular suitability	Shell-fit spike exists; proposed solver is unbuilt. Atomic trends do not establish chemical accuracy.
4 Chemistry Bond curves, geometry, channel search, barriers and surfaces	Molecular energetics; kinetics; detailed balance; polymer and reactor controls	Detailed proposal. Local search cannot guarantee discovery of every relevant pathway.
5 Materials Composition, microstructure and processing history	Mechanical and transport curves; hysteresis; phase behavior; reconstruction errors	Detailed proposal. Bounded history may omit decisive path dependence.
6 Continuum physics Gravity, fluids, heat, stress, radiation and electromagnetism	Convergence and conservation; orbit, convection, fracture, wave and engine benchmarks	Detailed proposal. Constitutive closure and coupling across scales remain untested.

Table 5. Physical phase map from PHASE0.md through PHASE6.md with proposed validation refinements. Criteria are research targets, not reported completions.

The phase order should be interpreted as a dependency graph rather than an obligation to finish an entire discipline before testing a connection. For example, a small thermally driven reaction system can test chemistry, heat exchange and logging before any planetary model exists. Such demonstrations reveal incompatible units, missing reverse processes and closure failures earlier than a full-world build.

The phrase every combination is calculable needs an operational domain: allowed species, conditions, approximation level, search budget and failure behavior. A simulator should report an unsupported state or unresolved pathway explicitly. Quietly returning no reaction would confuse model incompleteness with physical impossibility.

10 Roadmap from planets to institutions

The middle phases add environments, reproducing systems and agents that act through the physical world. Their common requirement is that higher-level summaries remain connected to resource flows and observable events. Their greater complexity also increases the risk of calibrating a preferred narrative instead of testing a mechanism. [S1]

Phase and intended mechanism	Required evidence	Primary research uncertainty
7 Planet and space Accretion, differentiation, climate, reservoirs and stellar summaries	Accretion/differentiation; closed water and carbon budgets; climate and LOD comparisons	Geological closure and long-time drift; seeded prior stellar output must be disclosed.
8 Life Templating, compartments, metabolism, heredity and ecosystems	Catalytic closure; persistence; heritable variation; competition; energetic and population agreement	Abiogenesis and complex bodies are open outcomes. Seeded replicators change the claim.
9 Agency and civilization Learned policies, bodies, tools, cooking and culture	Novel-resource tasks; tool performance; energy viability; cumulative learning; village comparisons	Long discovery chains and reward design may select the desired story.
10 Technology Experimentation, instruments, machinery, communication and computation	Energy/entropy closure; signal and grid tests; reproducible invention; nested replay	Hardware emergence and host-side nested simulation need an explicit equivalence test.
11 Society Communication, sanctions, learning, coalitions and institutions	Matched interventions; language/knowledge patterns; exposure-response; compression comparisons	Many mechanisms fit similar patterns. Human social categories need operational definitions.
12 Money and politics Ownership, obligations, credit, enforcement and diffusion	Consistent holdings; defaults and bank runs; real/nominal separation; conditional regime tests	Institutional outcomes are contingent. Accounting consistency does not validate behavior.

Table 6. Proposed phases 7 through 12. No implementation of these phases was identified in the inspected sim package.

Phase 7 permits disk composition seeded from an offline stellar calculation, and Phase 8 discusses a disclosed seeded-replicator fallback. Both can support useful experiments. They must be labeled as initialized boundary conditions rather than evidence that the entire preceding history emerged in the same realization. A continuous data record cannot by itself supply the missing causal history.

Phases 9 through 12 should evaluate distributions of possible histories, conditional on environments and agent rules. Fire, farming, computers, currencies or a particular political regime cannot be both guaranteed milestones and independent emergent predictions. Failed discovery, extinction and alternative institutions must remain admissible outcomes. The test protocol should specify what each outcome teaches before the seed is run.

11 Integration and unresolved dependencies

Phase	Roadmap scope	Proposed measurable gate
13 Infrastructure and externalities	Transport, industry, maintenance, waste, pollutants and harm	Closed material-flow accounting plus attribution of exposure and damage across owners and boundaries.
14 Perception and detail	Sensing, relevance, memory, information propagation and refinement	No privileged information access; preserved causal influence; consistent detail reconstruction with quantified errors.
15 Runtime	Data-oriented storage, tiered clocks, CPU/GPU work and replay	Measured workload scaling, memory/log growth, deterministic execution policy and fidelity-versus-cost curves.
16 Integrated loop	A persistent, playable history with endogenous innovation	An audited chain of transitions identifying every seed, shortcut, approximation and discovery, including failed emergence.

Table 7. Phases 13 through 16 have roadmap entries only. The right column supplies recommendations from this paper, not existing completed gates.

The seams ledger contains 18 cross-phase issues, of which 16 are marked open and two record wording fixes. These include rotating-frame physics, rheology, nucleation, gravity and expansion, stellar yields, logging of boundary conditions, templating, compartments, photochemistry, rigid bodies, dynamic electromagnetism, nested replay and ownership transfer. Several later phase documents assume that these upstream interfaces will exist. They therefore cannot be implemented independently merely by following phase numbers. [S3: seams]

Perception and optimization are cross-cutting dependencies, although they appear as phases 14 and 15. Quantum probes, geological jumps, population summaries and social memory already rely on their decisions. Their contracts should be established early and improved through measured examples. A late performance pass cannot recover information discarded by an invalid closure.

Three tensions need explicit resolution. First, claims of exact equality after coarse-graining conflict with genuinely lossy summaries unless they are restricted to particular observables or domains. Second, some phase gates require specific historical developments while the project also promises open outcomes. Third, host-side execution of a nested simulator is not automatically equivalent to computation physically carried out by an in-world computer. Each tension can be turned into an experiment with declared assumptions.

A phase completion record should identify the source revision, calibration dataset, held-out tests, known failure domain, measured costs and the next interface demonstrated. Checkmarks alone are insufficient. This record would make the roadmap a cumulative research program in which a failed validation can revise the model without requiring the entire project narrative to remain intact.

12 Experimental design and uncertainty

Verification before empirical comparison

The next experiments should establish that each stochastic kernel samples the model it claims to sample. Test reversible systems from both sides of equilibrium, including zero-reactant and fully occupied states. Compare small reaction networks against an analytic stationary distribution or a trusted stochastic reference. Separate errors in reaction enumeration, rates and event sampling. A time-step refinement study is meaningful only after physical time and per-step rates have a consistent relationship. [8]

Conservation tests should account for all modeled reservoirs and boundary fluxes, rather than only algebraic identities derived from the same state variable. Tests should also expose forbidden transitions, negative populations, inconsistent unit conversions and changes caused by event ordering. Replay corruption and truncation tests remain valuable because they establish integrity of the experimental record. [S5]

Calibration and independent validation

Before adjusting a parameter, declare which observables calibrate it, which are held out and which conditions lie outside the model's domain. Nuclear anchors and freeze-out/yield bands used to tune constants cannot simultaneously serve as independent evidence for those constants. Held-out isotope systems, temperature regimes, compositions and initial populations are more informative than rerunning the fitted cases.

A proposed initial study would use fixed calibration data, at least 30 held-out seeds, and additional seeds until uncertainty intervals stabilize. Thirty is a planning starting point, not a guarantee of power. Held-out seeds assess stochastic robustness; independent empirical validation still requires external observations or trusted reference solutions. Rare-event designs depend on expected counts and confidence limits. Stream overlap must be corrected or accounted for before independent-sample calculations are applied.

Propagation across scales

Uncertainty includes stochastic sampling, parameter uncertainty, numerical discretization, missing mechanisms and model reduction. These components should be reported separately. A compact organizing relation is $y = F(\theta, s, h, C)$, where θ denotes parameters, s the random seed, h the numerical resolution and C the chosen closure. Varying only s measures neither parameter sensitivity nor structural error.

For each compression level, compare detailed and reduced runs under matched conditions. Report bias in primary observables, distributional distances, conservation residuals and rare-event rates as functions of computational cost. Correlated errors can accumulate across interfaces, so validating each layer in isolation does not bound the error of the combined world. Cross-layer interventions are needed to identify where a result becomes sensitive to a closure. [6]

Publish negative results and the settings that produce them. A model of emergence is especially vulnerable to selecting successful seeds or adjusting rules until a desired transition appears. Predeclared observables and all-seed reporting make it possible to distinguish a reproducible mechanism from a curated history.

13 Priority experiments and decision criteria

The following sequence is proposed work. It is ordered to remove uncertainties that would otherwise propagate into every later phase. Completion requires the associated evidence, not simply an implementation or a visually plausible trajectory.

Priority experiment	Protocol and primary observable	Criterion for advancing
1 Reversible reaction kernels	Product-only nuclear initial states; neutral and ionized atom states; multiple fixed temperatures; reference stationary distributions	All intended reverse pathways exist; the same equilibrium is reached from both sides within predeclared statistical error.
2 Random-stream identity	Distinct species/events, reordered independent work and parallel scheduling; audit counter uniqueness and correlation	Unique documented keys for independent draws; stable replay; passing independence and scheduling checks.
3 Frozen physical baseline	Dimensional rate mapping; conservation budgets; resolution and step refinement; held-out nuclei and atomic cases	A published error budget and domain of validity; calibration and prediction results reported separately.
4 Measured runtime model	Integrated engine at several population sizes and observation patterns; repeated timings and memory/log accounting	A defensible cost curve for the actual implementation at the required fidelity.
5 One coupled physical experiment	A small reaction system exchanging heat with a reservoir, with detailed and reduced representations	Mass/energy budgets close; kinetic and thermal observables agree with reference within declared tolerances.
6 Open-ended extension	Add compartments after validated chemistry, or tool use after validated mechanics, sensing and learning; test controls and new environments	The measured new behavior depends on the proposed mechanism and survives held-out conditions.

Table 8. Recommended experimental order. Numerical tolerances must be chosen for each observable and benchmark before evaluating the results.

Every experiment should archive its model version, parameter values, inputs, initial state, random-stream schema, raw outputs, analysis procedure and failure cases. Changes to physics or random-stream identity require a new baseline. Historic files should remain replayable with their original code or be explicitly marked as requiring an archived environment.

The first coupled experiment is deliberately much narrower than a planet. It tests the project's central proposition: that independently checked mechanisms can be composed without losing accounting or explanatory value. Success would justify a larger coupled system. Failure would locate an interface problem before it is hidden by biological or social complexity.

14 Research questions for life and social emergence

From reaction networks to reproducing systems

An autocatalytic network is an intermediate chemical property. It does not by itself demonstrate heredity, selectable variation, compartment persistence or open-ended evolution. Research on autocatalytic sets explicitly separates catalytic closure from a complete account of life's origin. Phase 8 should therefore report these properties individually and use non-catalytic, non-templating and resource-limited controls. [9]

A falsifiable question is whether templating increases lineage persistence and heritable variation under a fixed resource flux compared with a matched network without templating. A second is whether compartments change competition or extinction rates after controlling for total material and energy. These comparisons need direct chemical and population observables, not merely a detector that labels a pattern as life.

Digital evolution has shown complex functions emerging in engineered populations of replicating programs. It supports controlled studies of novelty and historical contingency, while supplying no proof that a physical chemistry model will generate life or human-like intelligence. Any seeded replicator or designed reward must be declared as part of the experiment. [10]

From agents to knowledge and institutions

For agency, separate what an agent can sense, remember, infer and physically do. Tool-use success should depend on the modeled mechanics and resource cost. Research claims should compare learners with restricted memory, communication or experimentation and quantify task performance, energy expenditure and transfer to unfamiliar environments. A novelty measure must distinguish new behavior from relabeling an existing action.

For cumulative knowledge, test whether communication and retained records improve performance across generations under matched ecological conditions. For economic organization, test whether alternative ownership, credit and enforcement rules change exchange, defaults, production and inequality. For institutions, use interventions on mechanisms and report the range of outcomes across seeds. ODD-style documentation can expose scheduling and initialization choices that otherwise resemble emergent social laws. [7]

One proposed institutional experiment compares otherwise matched resource-sharing communities with different communication reliability and sanction costs. Resource persistence, enforcement expenditure and outcome distributions would distinguish successful coordination from a prescribed institutional label. Initial endowments, agent capabilities and external shocks must be matched, and failures retained.

The proposed social variables are operational abstractions. They do not yet justify claims that mental health follows directly from a small neurochemical state, that a named culture has a fixed mechanism, or that a political system has a universal performance ordering. Human-facing interpretations require independent empirical work and clear limits on what the agents represent. The value of the simulator at this stage would be testing consequences of explicit assumptions, not assigning scientific inevitability to a preferred society.

15 Discussion and research contribution

SimWorld's roadmap is most useful as a program for testing interfaces between effective models. Its breadth creates a demanding requirement: every upward summary must preserve the consequences that the next level relies on, while every downward action must be paid for in the relevant physical or informational budget. A coherent history is valuable only when the meanings of its events remain stable across those transformations.

The existing build establishes a practical starting point. It executes a cooling sequence, produces aggregate bound states, records stochastic causes and reproduces its output. The fresh tests and replay provide direct evidence for these engineering properties. The source audit also demonstrates why documentation and regression success are insufficient measures of scientific validity: explicit stability filters shape outcomes, reverse reactions can disappear from enumeration, and a neutral state can remain fixed when the model's own equilibrium comparison predicts ionization.

These limits do not make the reduced model uninformative. They define which experiments it can currently support and which claims must wait. The immediate scientific opportunity is to use its reproducibility machinery to isolate and correct kinetic assumptions, establish independent benchmarks and quantify the consequences of approximation. Those steps are more informative than extending the timeline before the basic kernels have a validated domain.

The project should also distinguish three senses of emergence. **Computational emergence** is a pattern generated by specified local rules. **Robust emergence** persists across reasonable parameter, resolution and initialization changes. **Empirical adequacy** is agreement with the intended real system on independent evidence. A phenomenon can satisfy the first definition without the second or third. This distinction makes success measurable while preserving room for useful exploratory worlds.

The long-term claim of an unscripted route from a quantum seed to civilization remains open. The phase documents already introduce effective fields, fitted forces, reduced chemistry, constitutive material rules, learning mechanisms and seeded boundary conditions. These are legitimate modeling tools when disclosed and tested. The program should measure what causal information they preserve rather than describe the whole stack as a deduction from a minimal set of universal constants.

Conclusion

The current evidence supports a reproducible reduced simulation and a detailed agenda for its extension. It does not yet support quantitative cosmological fidelity or completed higher-level emergence. Progress toward a research-grade platform should be judged by reversible and benchmarked kernels, independent validation, explicit cross-scale error contracts and reproducible experiments that can fail. A small validated chain of coupled mechanisms would be a stronger next contribution than an unvalidated realization of the entire roadmap.

Appendix A Reproduction and data provenance

The accompanying archive preserves the source snapshot used for this paper, including phase documents, implementation, tests, prototype records and archived M3/M4 outputs. The research directory adds the fresh M4 run, diagnostics, full-suite report, literature metadata, manuscript and source manifest. Source paths in this paper refer to paths within that snapshot. [S7]

Environment and commands

Engine verification used Python 3.11.9, NumPy 2.4.6 and pytest 9.1.1. The platform API reported Windows-10-10.0.26300-SP0; the build specification calls the machine Windows 11. These labels are recorded without inferring operating-system identity from the API string alone. Execute the following from the source directory with the stated Python environment:

```
python -m pip install -r requirements.txt
python -m pytest -q
python -m sim.demo --seed 42 --ticks 3400 --cells 8
  --probe-cells 2,3 --out reproduced_m4
python -m sim.replay reproduced_m4
```

The two displayed demo lines are one shell command; join them with a space. Installing dependencies changes the selected Python environment, so an isolated environment is appropriate for reproduction.

Identity checks

The fresh summary matched _m4run/summary.json exactly. Both hash sequences matched. The following fingerprints permit exact comparison without treating rounded paper values as the reproducibility target:

Final state SHA-256
`fe8e094a9753b27c89fc0dffd1905f8de4be`
`45263158d5712618ddd5f6ded1a0`

Decompressed event-log SHA-256
`15e50cd6adbc6cc0f01360ebd6fc4a8a97a7`
`06e0cb459f0fd218abeafba777f3`

Each displayed fingerprint is one 64-character hexadecimal string; remove the visual line break.

The fresh run took 5.550 seconds and its replay 5.547 seconds while pytest was also active. These observations under concurrent load have no repeated-trial confidence interval. The full-suite JUnit report and text output preserve the 65-pass result. The older nightly report is historical evidence only. To repeat the diagnostic cases from the archive root, use:

```
python research/audit_diagnostics.py --source-root source
  --output-dir reproduced_audit
```

Join the two displayed lines into one command. The output directory must be new, preserving archived evidence.

Scope of reproducibility

Exact equality is supported for this source and environment. Cross-platform floating-point equivalence, independence of stochastic streams, and equivalence under different scheduling policies require separate evidence. The source_manifest.json file identifies every included project file by byte size and SHA-256, while the original physics_build remains 0.1. The evidence package enables scrutiny of both successful regression tests and the targeted counterexamples.

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Project sources and evidence index

[S1] SimWorld roadmap and phase specifications. ROADMAP.md; PHASE0.md through PHASE12.md. Snapshot 29 October 2026. Phase mechanisms and proposed gates are identified by phase number in Sections 9-11 of this paper. Phases 13-16 are described in ROADMAP.md only.

[S2] SimWorld first build contract. BUILD0.md. Relevant sections: Environment; What ships; Decisions; Physics build 0.1; Engine architecture; Physics flow per tick; verification and milestone acceptance criteria.

[S3] SimWorld specification records. spec/constants.md; spec/decisions.md; spec/interfaces.md; spec/surprises.md; spec/seams.md. Decisions and interfaces carry the physics_build 0.1 freeze date of 28 October 2026. The surprises and seams ledgers are records of known limitations and planned changes.

[S4] SimWorld implementation snapshot. sim/constants.py and fields.py define inputs; lattice.py and condense.py prescribe cooling and population targets; quanta.py defines record storage; weak.py and bind.py implement flavor conversion and nucleon formation; nucbind.py and network.py define effective nuclei and reactions; atoms.py defines capture; rng.py, log.py, state.py and replay.py define repeatability; jump.py and demo.py coordinate the run.

[S5] SimWorld verification suite. tests/test_rng.py; test_log.py; test_replay.py; test_m2_physics.py; test_m2_jump.py; test_m3_nuclei.py; test_m4_atoms.py; test_physics_bands.py; physics_bands.toml; tests/oracle/. The frozen anchors and acceptance criteria are part of the model-development evidence, not wholly independent physical validation.

[S6] Archived runs and prototype reports. _m3run/summary.json, run.jsonl.gz and hashes.json; _m4run equivalents; spikes/a_quant_cost/report.json and run.py; spikes/b_shells/report.json, run.py, shells.py, oracle_ips.json and errors.csv. Reported measurements are attributed to these individual workloads.

[S7] Fresh audit evidence generated 29 October 2026. research/audit_tests.txt and audit_tests.xml; audit_diagnostics.py and audit_diagnostics.json; fresh_m4/; implementation_audit.txt; roadmap_audit.txt; literature.json; source_manifest.json. Reproduction is described in Appendix A. The supplementary archive also contains the editable manuscript and PDF builder.

Evidence locations for the principal findings

Finding	Source location
Dictionary/list quant storage	sim/quanta.py, QuantStore.__init__ and add_superposed
Reverse nuclear channel omissions and formula	sim/network.py, _rates; sim/bind.py, channel
Fully neutral absorbing state	sim/atoms.py, step; audit_diagnostics.json, atomic_absorbing_diagnostic
Random-prefix reuse	sim/log.py, EventContext.draws; audit_diagnostics.json, rng_reuse_diagnostic
Fresh equality and replay	audit_diagnostics.json, snapshot_comparison and runs; fresh_m4/hashes.json

Table 9. Function-level locators remain usable in the archived source even when line numbers change in later development.